

Emily Choong

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Skills

Tools: Python, SQL, R, Excel, Tableau, AWS (SageMaker, Redshift, Bedrock), Git, PowerPoint, Claude

Skills: Machine Learning (XGBoost, Random Forest), NLP, LLMs & RAG, AI Agents, Prompt Engineering, Data Engineering & Pipelines, Data Classification & Normalisation, Statistical Modelling, Model Evaluation

Education

University of Bristol

Bristol, England

BSc Data Science with a Year in Industry: On track for 1st class honours

2023-2027

Relevant Units: Advanced Linear Modelling and Classification, Probability & Statistics, Algorithms and Machine Learning, Financial Mathematics, Time Series and Stochastic Processes

Shrewsbury School

Shrewsbury, England

*A*A*A*A*: Math, Further Math, Chemistry, Physics*

2022-2024

Work Experience & Projects

Data Science Intern (Year in Industry)

Sep 2025 - 2026

Pacific Life Re, London

Cause of Claim Classifier - using NLP

- Built an NLP pipeline using TF-IDF and a hierarchical Random Forest classifier to categorise free-text insurance claims into a three-level taxonomy of 256 labels, achieving 95% accuracy and automating classification for over 90% of records.
- Designed a human-in-the-loop workflow to review low-confidence classifications and unseen claim reasons, ensuring appropriate oversight of automated decisions.
- Led the integration of LLMs into the classification pipeline, conducting risk analysis and designing the end-to-end data flow for securely connecting claim data to an LLM via AWS Bedrock API. Collaborated with data engineers, risk and compliance teams to develop a scalable and responsible approach to deploying AI in a core business workflow.

Data Schema AI agent

- Built an AI agent to make a complex database more accessible to new users, reducing time required to understand and navigate unfamiliar data.
- Developed a Python pipeline to extract and structure metadata from core tables across 5 schemas (~2 billion rows), creating a machine-readable JSON knowledge base containing fields, data ranges, examples and join keys.
- Tested and iterated on the agent using user feedback, refining prompts and custom instructions to improve response precision, consistency and readability.

UW Normalisation (structuring underwriting data for risk & pricing)

- Transformed 500,000+ unstructured underwriting Q&A pairs into 9 interpretable risk markers using text embeddings and K-means clustering, standardising 2.2M records into a structured dataset.
- Built and evaluated a decision-tree model (held-out AUC 0.977) to validate the markers, identifying where they differentiated declined from standard risks and predicted pricing loadings.
- Following the pilot, explored using LLMs to scale the transformation into structured risk markers, investigating domain context, output testing, monitoring and iteration.
- Communicated findings to non-technical underwriting and actuarial teams, translating statistical results into clear business insights to build commercial buy-in.

Survival Risk Modelling from UK Primary Care Data

- Built and cleaned a large-scale patient dataset (~2M individuals, 30+ tables) from linked EHR sources, developing a reproducible Python pipeline on AWS (SageMaker, Redshift) to resolve missingness, inconsistencies and schema misalignment and create a modelling-ready dataset.
- Transformed complex clinical data into interpretable predictive features, applying research-backed banding and MICE imputation to address ~50% missingness, with sensitivity analysis to test the robustness of assumptions.
- Developed and evaluated Cox proportional hazards and XGBoost survival models to identify drivers of time-to-event outcomes, validating model assumptions and distinguishing robust relationships from potential confounding.

Product Intern - AI Biotech startup

Jul - Aug 2025

Dr KuDos Lab, Hong Kong

- Worked on an AI-powered cosmetic surgery doctor avatar, exploring how to fine tune an LLM specialised in cosmetic procedures and surgery, which designing the underlying data structure and RAG architecture.
- Redesigned the onboarding flow to shorten the path to a new user's first meaningful action, reducing onboarding friction and supporting customer acquisition
- Identified features to drive long-term engagement and retention, including a gamified points system and a patient diary; the diary was approved by the founder for the product roadmap

Extracurriculars & Interests

Badminton Club Lead – *Pacific Life Re*; organised regular matches and monthly social events for cross-team bonding.

Captain of Badminton team – *Dunman High School*; Build communication and team spirit within the team.

Languages: English (Native), Mandarin (Proficient)

Interests: Hot yoga, Badminton, Running – participated in the Bath Half Marathon (2025), Reading, Music – Grade 8 Piano & Violin